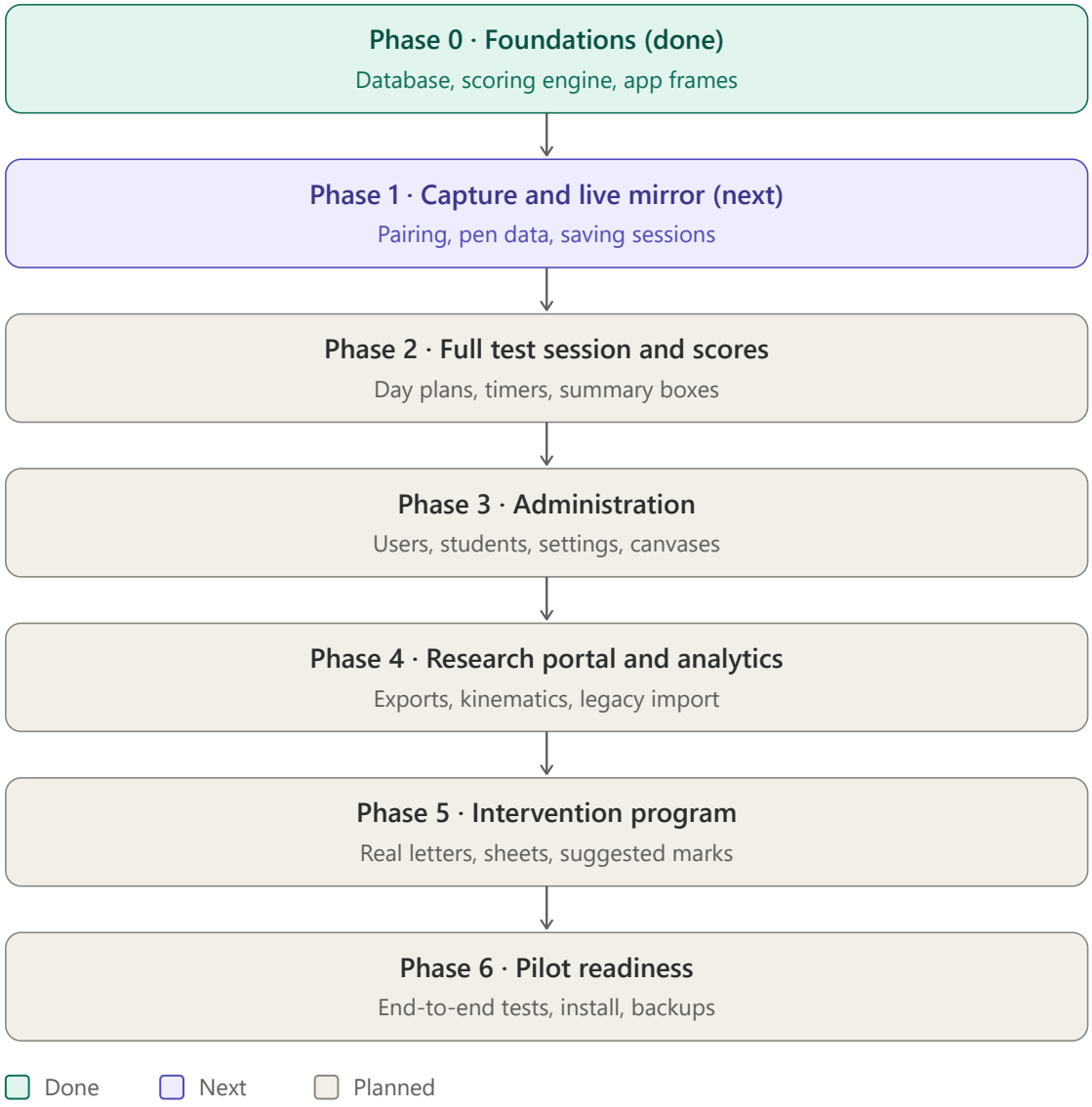
ILT Research System: Roadmap and Architecture

29 October 2026 · Status: Phase 0 complete (commit e83beae, CI passing) · Sources: FRS v3.5, Architecture v5.6, DDD v1.6, Clinical Decisions Register

1. Roadmap

Phase 1 follows the coding-readiness plan. Phases 2–6 are a proposed order built from the FRS and the Clinical Decisions Register; they are not yet fixed in the specifications.



What each phase delivers

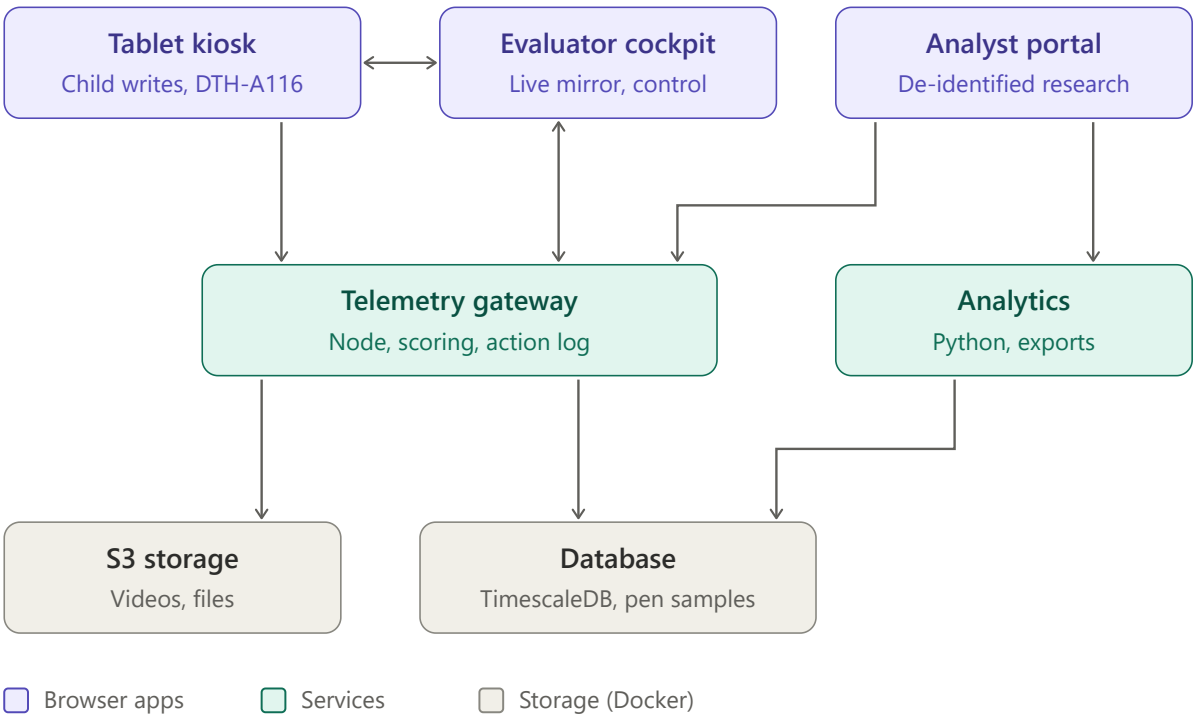
- **Phase 0 · Foundations (done):** monorepo and CI; database schema with roles and the settings-scope rule; seed data (settings, ILT and notebook canvases, day plans, Regions, dev accounts, Hebrew videos); scoring engine with the worked examples; pen-data format with error check; Hebrew/English translations; gateway health and catalogue; kiosk shows the ILT page at true size.
- **Phase 1 · Capture and live mirror:** the cockpit pairs with a tablet using a short code and controls the session; the kiosk captures every DTH-A116 pen sample and sends it to the gateway in batches; the evaluator sees strokes live; sessions, blocks and samples are saved, and nothing is lost if the connection drops.
- **Phase 2 · Full test session and scores:** the Day 1, Day 2 and Day 14 flow with the warm-up block, intro video, inactivity timers and director PIN; each block scored when it finishes; summary boxes with the Replay button. After this phase a real test can run start to finish.
- **Phase 3 · Administration:** System Administrator screens for users and roles, students, Domains and Regions, and settings with their overrides; the canvas library with reference recordings (Accept / Clear); action-log search.
- **Phase 4 · Research portal and analytics:** the de-identified analyst portal; pen-movement (kinematic) analysis; exports; import of the old application's data.
- **Phase 5 · Intervention program:** real letters א ת ח ק ה, the letter sheets, the three writing levels, and suggested marks for the evaluator to confirm.
- **Phase 6 · Pilot readiness:** end-to-end tests on the DTH-A116, installation on the school PCs, backups, and the ethics checks needed before real children are tested.

Open items

- Notebook page line spacing is provisional and needs checking against a real school notebook.
- Dot positions for the intervention letters still need to be measured from the letter sheets.
- Final Phase 1 checks need the DTH-A116 connected to the development PC.

2. Architecture

Simplified from Architecture v5.6 §2. The three apps run in the browser; the services and storage run on the site's ILT server in Docker.



How the parts connect

- **Tablet ↔ cockpit, directly:** the evaluator sees the child's strokes with almost no delay, and the cockpit's start, pause and next-step commands reach the tablet the same way.
- **Tablet → gateway:** in parallel, the tablet sends every pen sample to the gateway in batches. This is the saved copy; the live mirror is for watching only.
- **Cockpit ↔ gateway:** students, sessions and scores.
- **Gateway → storage:** pen samples, scores and the action log go into the database; videos and files go into S3 storage (SeaweedFS).
- **Analyst portal:** reads only the de-identified research views through the gateway, and gets exports and pen-movement analysis from the analytics service, which reads the database.

Not shown in the drawing

- Analytics also reads and writes S3 storage (for exports).
- The optional desk camera goes through the go2rtc camera relay to the cockpit and S3 storage.
- A web server (Caddy) serves the three apps over HTTPS.
- The analyst portal reaches the school's server over a VPN.

Phase 1 builds the left half of this drawing: the tablet-to-cockpit link and the tablet-to-gateway saving path.